DATA ANALYSIS REPORT

Retailer Sales Analysis

Sales performance, KPIs, trends and customer patterns

PERIOD	1 May 2026 – 30 Jun 2026
DATASET	500 orders	26 columns
TOOLS	Python	pandas	matplotlib

1. Dataset Overview & Objective

Item	Detail
Source file	Retail_Sales_Performance_Analysis_Final.xlsx (retailer sales data)
Size	500 orders and 26 columns
Period covered	1 May 2026 to 30 Jun 2026 (two months)
Objective	Clean the retailer sales data, calculate core business KPIs, identify sales trends and customer patterns, and present findings with charts and recommendations.

2. Data Cleaning Summary

The raw Excel file was left unchanged. All cleaning was done on a copy in the notebook, and the cleaned data was saved separately as **cleaned_retail_sales.csv**.

Check	What was done	Result
Missing values	Counted nulls in every column	None found
Duplicate rows	Checked for fully duplicated rows	None found
Repeated Order IDs	Checked if any Order ID appears more than once	None found
Negative quantity	Checked for quantity below 0	None found
Zero / negative sales	Checked for sales of 0 or below	None found
Delivery Status & Product Category	Listed unique values to look for spelling variations	Values were consistent
Order Date	Checked the date column and its range	Stored as Excel serial numbers; converted to real dates

Date fix. Order Date came from Excel as serial numbers instead of real dates. It was converted with pd.to_datetime(), giving a proper range of 1 May 2026 to 30 Jun 2026. This was necessary for the weekly trend analysis.

Decision. The date format was the only problem. The data had no missing values, duplicates or invalid entries, so no rows were deleted or filled in.

3. KPI Summary

TOTAL REVENUE
Rs. 1.25 Cr
12,482,584.35 after discounts

TOTAL ORDERS
500
unique orders

AVG. ORDER VALUE
Rs. 24,965
revenue / orders

UNITS SOLD
1,072
sum of quantities

TOP CATEGORY
Desert Cooler
highest revenue

TOP PRODUCT
Symphony Diet 35T
highest revenue

All 500 orders are included in the KPIs. 19 orders (about 3.8% of orders, 4.46 lakh in sales) were cancelled and 38 are still in transit, so revenue may be slightly overstated.

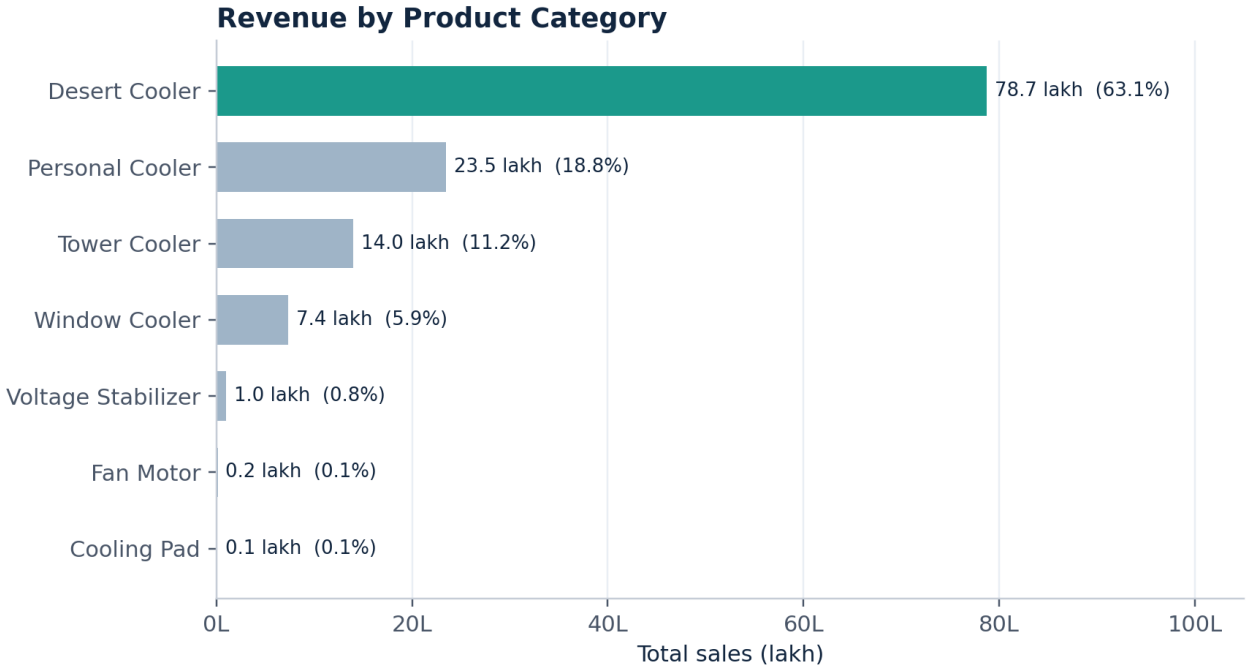
Delivery status breakdown

Status	Orders	Share of orders	Sales (Rs.)
Delivered	443	88.6%	11,059,071.65
In Transit	38	7.6%	977,199.85
Cancelled	19	3.8%	446,312.85

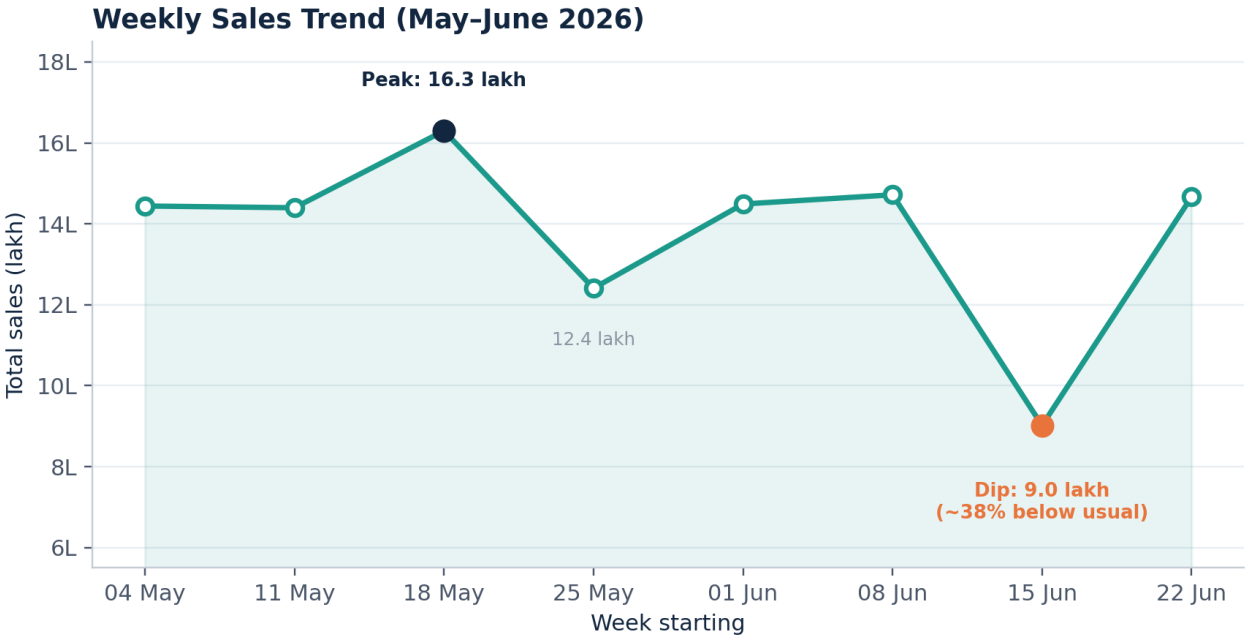
4. Key Findings

1. **Category:** Desert Cooler brings in about **63%** of revenue; the top three categories (Desert, Personal, Tower) give about **93%**. Fan Motor, Cooling Pad and Voltage Stabilizer make up only about 1%.
2. **City:** Ghaziabad is the top city with about **28%** of revenue, and the top five cities together contribute about **85%**.
3. **Payment mode:** UPI is most used (about **47%** of orders), then cash (29%) and card (20%). Bank transfer is rare (about 4%).
4. **Delivery status:** About **89%** of orders were delivered, 7.6% are in transit and 3.8% were cancelled.
5. **Weekly trend:** Weekly sales were steady at about 14–16 lakh, peaking in the week of 18–24 May (about 16.3 lakh). The week of 15–21 June dipped to about 9.0 lakh, roughly **38%** below the usual level. The data doesn't explain why.
6. **Customers:** Orders are spread almost evenly across Medium (174), Low (171) and High Value (155) segments, so no single segment dominates.

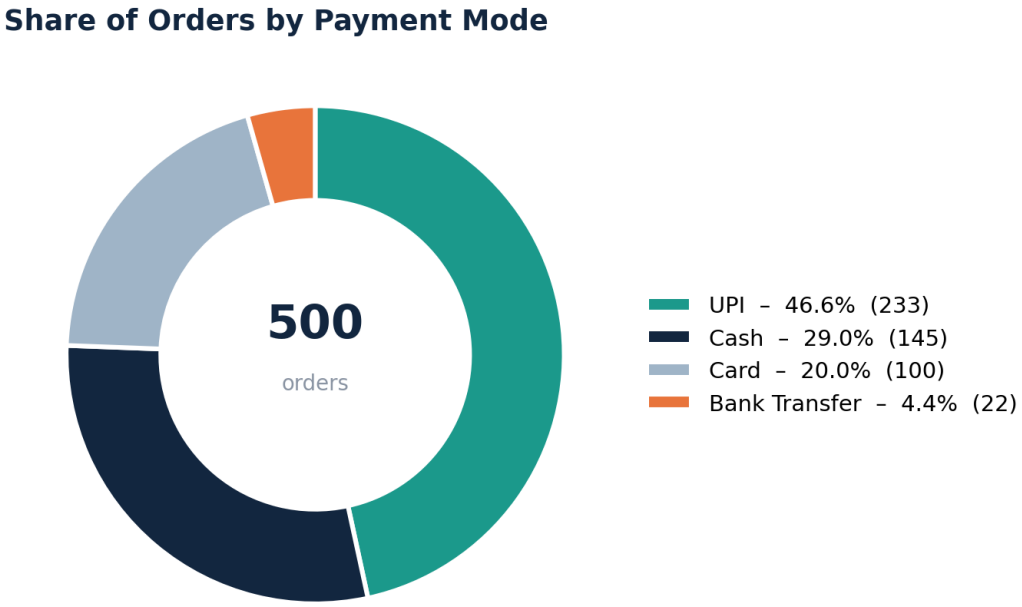
5. Visualizations



Revenue by Product Category. Desert Cooler earns far more than any other category, about 63% of total revenue (around 78.7 lakh). Personal Cooler (18.8%) and Tower Cooler (11.2%) come next, and Window Cooler is a distant fourth. Voltage Stabilizer, Fan Motor and Cooling Pad earn so little that their bars are barely visible. The business depends mostly on Desert Coolers.



Weekly Sales Trend. Weekly sales were mostly steady at around 14–15 lakh, peaking in the week starting 18 May at about 16.3 lakh. There were two dips: a small one in the week starting 25 May (about 12.4 lakh) and a sharp one in the week starting 15 June (about 9.0 lakh, roughly 38% below the usual level). Sales recovered the following week. The first and last weeks were left out because they only cover part of a week.



Share of Orders by Payment Mode. UPI is the most used mode, covering about 47% of orders, followed by cash (29%) and card (20%). Bank transfer is rare at about 4%. Digital payments (UPI and card) make up roughly two-thirds of orders, while cash still makes up a significant share.

Top 5 cities by revenue

City	Revenue (Rs.)	Share
Ghaziabad	3,513,527.20	28.1%
Noida	2,267,410.20	18.2%
Greater Noida	1,965,270.25	15.7%
Meerut	1,573,718.55	12.6%
Muradabad	1,289,030.70	10.3%

6. Customer Insights

Customer segments. Orders are spread almost evenly across Medium (174), Low (171) and High Value (155) segments. The business does not depend on one type of customer.

Payment behavior. UPI is preferred (about 47%), followed by cash (29%) and card (20%). UPI and card together cover roughly two-thirds of orders, so most customers prefer digital payments. Cash still accounts for almost 3 in 10 orders, meaning many customers like to pay on delivery or in person.

Customer base. The 500 orders came from 167 unique customers, about 3 orders per customer. At least 333 orders (about 67%) were repeat purchases, so keeping existing customers happy matters as much as finding new ones.

Calculation: 167 orders are first orders (one per customer), so 500 − 167 = 333 repeat orders, and 333 ÷ 500 = 67%.

7. Recommendations

1. **Protect and grow the Desert Cooler line.**
Desert Coolers bring in about 63% of revenue, so keeping them well stocked before and during the season matters most.

2. **Review the low-earning categories.**
Fan Motor, Cooling Pad and Voltage Stabilizer make up only about 1% of revenue. Test bundling them with coolers as add-ons, or reduce the stock held for them.

3. **Reduce dependence on a few cities.**
Ghaziabad gives about 28% of revenue and the top five cities about 85%. Test marketing in other cities to spread the risk.

4. **Look into cancelled and in-transit orders.**
About 3.8% of orders were cancelled and 7.6% are still in transit. Finding out why would help protect revenue.

5. **Investigate the June dip.**
Sales in the week of 15 June fell about 38% below the usual level. Checking stock, delivery or local events for that week could show whether it can be avoided next time.

8. Limitations & Assumptions

- **Cancelled and in-transit orders are included.** All 500 orders count toward revenue, including 19 cancelled and 38 in-transit orders, so revenue may be slightly overstated.
- **Short time period.** The data covers only two months (May–June 2026), so nothing can be said about seasonal patterns across a full year.
- **Partial weeks removed.** The first and last weeks are partial, so they were left out of the weekly trend.
- **Causes are not known.** The data shows what happened (e.g. the June dip) but not why. Recommendations are suggestions to investigate, not proven causes.
- **Customer analysis basis.** Insights come from customer segments, payment mode and order counts per customer: 167 customers and 500 orders, about 67% repeat purchases.